

UCI: Division of Continuing Education
Accelerated Certificate Program (ACP): Data Science
Introduction to Python Programming
 I&C SCI_X426.58:

Homework#2

Date Given: April 4, 2024

Due Date: April 5, 2024

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There are 7 problems in this homework assignment. Write a program in Python to solve these problems.

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Problem#1

Determine the value of i^i as a real number, where $i = \sqrt{-1}$.

In other words, compute i to the power of i , where $i = \sqrt{-1}$.

Hint: Use Euler's formula: $e^{ix} = \cos x + i \sin x$

If $x = \frac{\pi}{2}$ $e^{i\frac{\pi}{2}} = i$

Problem#2

If two adjacent sides of a regular, 6-sided die have the values a and b when viewed side-on and read left-to-right, the value on the top of the die is given by $3(a^3b - ab^3) \bmod 7$.

Determine the value on the top of the die if (a) $a = 2, b = 6$, (b) $a = 3, b = 5$.

Problem#3

How many times must a sheet of paper (thickness, $t = 0.1$ mm but otherwise any size required) be folded to reach the moon (distance from Earth, $d = 384,400$ km)?

Please note that after every fold of the paper, the thickness of the paper will double.

Fold	0	1	2	3	4	5
Thickness - mm	0.1 mm	0.2	0.4	0.8	0.16	0.32

Therefore, the thickness of the paper after n^{th} fold will be $2^n t$.

Problem#4

Explain why the following expression does not evaluate to 100.

```
>>> 10^2
8
```

Hint: Refer to the Python documentation for `{bitwise} operators`.

Problem#5

Slice the string `s = 'seehemewe'` to produce the following substrings:

- (a) `'see'`
- (b) `'he'`
- (c) `'me'`
- (d) `'we'`
- (e) `'hem'`
- (f) `'meh'`
- (g) `'wee'`

Problem#6

Write a single-line expression for determining if a string is a palindrome (reads the same forwards as backwards).

For example, word **'banana'** is not a palindrome. And word **'deified'** is palindrome.
Other examples: **noon, civic, racecar, level, and mom.**

Problem#7

Consider the following (incorrect) tests to see if string `s` has one of two values. Explain how these statements are interpreted by Python and give a correct alternative.

```
>>> s = 'eggs'
>>> s == ('eggs' or 'ham')
True

>>> s == ('ham' or 'eggs')
False
```